

BAIO Preparatory Stage Curriculum (Grades 3-5)

Bharat AI Olympiad - CBSE CTAI 2026-27 Aligned Framework

1. EXAM PATTERN & MARKING SCHEME

Total Questions: 35 | Duration: 60 minutes | Total Marks: 40 Marks

Section A: CT & Logical Reasoning (10 Questions, 10 Marks)

Section B: AI & Technology Concepts (10 Questions, 10 Marks)

Section C: Everyday AI Around Us (10 Questions, 10 Marks)

Section D: Innovation Arena (5 Questions, 10 Marks - 2 Marks each)

There is no negative marking.

2. GRADE 3 SYLLABUS & OUTCOMES

Logical Units: Algorithmic sequencing, pattern recognition, decomposition of 2-3 step problems, basic 3D spatial reasoning (cube viewpoints, mirror images), block-based coding concepts.

AI Units: Introduction to AI - voice assistants, face recognition, YouTube recommendations, autocomplete. How AI learns from examples vs how regular programs work.

Sample Question: A computer is given the rule: IF number is even, PRINT EVEN. IF number is odd, PRINT ODD. For 7, what does it print?

3. GRADE 4 SYLLABUS & OUTCOMES

Logical Units: Flowcharts with YES/NO decision diamonds, input-output function rules, multi-step decomposition, mirror images and symmetry, grid movements, loop patterns.

AI Units: Training data and supervised learning. Why spam filters work. How IRCTC AI detects fraud bots. Differences between AI and automation.

Sample Question: An AI recommends "Study more" IF exam is tomorrow AND score < 70. Rohan's exam is tomorrow and his score is 65. What does the AI recommend?

4. GRADE 5 SYLLABUS & OUTCOMES

Logical Units: Decision trees with 2^n path counting, multi-rule patterns, pictorial decomposition, water images vs mirror images, combined transformations, Caesar cipher basics.

AI Units: All three AI learning types - supervised, unsupervised, reinforcement. Deep learning basics. AI4Bharat and IndicTrans2. AI bias - why training data matters.

Sample Question: A table shows rainfall and crop yield over 4 years. More rain = more yield. What AI technique models this relationship? Predict yield for 300mm rain.